



STAGE 5.1-5.3 MATHEMATICS

Year 9 Assessment Task 2

May 10, 2012

General instructions

- Working time – 50 minutes.
- Write your answers the supplied A4 lined paper.
- Marks may be deducted for careless or poorly arranged work.
- Commence each new question on a new page.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the sheets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 9M1 – Mr Ireland/Mr Berry
- ☐ 9M2 – Mr Lam
- ☐ 9M3 – Ms Ziazaris/Mr Lowe
- ☐ 9M4 – Mr Lin/Mr Lucas
- ☐ 9M5 – Mr Weiss/Mr Berry

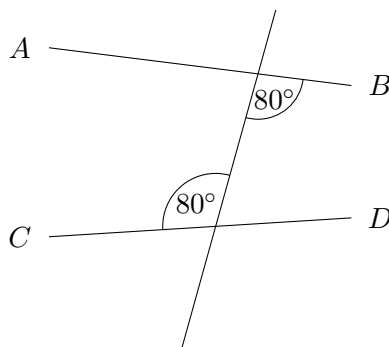
NAME: # SHEETS USED:

Marker's use only.

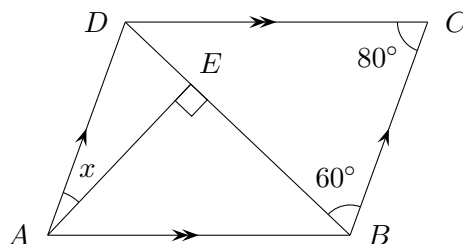
QUESTION	1	2	3	4	Total	%
MARKS	$\overline{12}$	$\overline{17}$	$\overline{13}$	$\overline{11}$	$\overline{53}$	

Question 1 (12 Marks)	Commence a NEW page.	Marks
(a) Write 19.449 correct to		
i. 2 significant figures.		1
ii. 2 decimal places.		1
(b) Evaluate $\sqrt[3]{13^2 - (12.8)^2}$ correct to 2 decimal places.		2
(c) What is the simple interest on \$850 for five years at 12% p.a.?		2
(d) What is the size of each exterior angle of a regular octagon?		1
(e) Given $R = \sqrt{\frac{V}{\pi h}}$, find R when $V = 500$ and $h = 12$, correct to two decimal places.		2
(f) Write the basic numeral for 2.15×10^{-4} .		1
(g) Simplify $(4.3 \times 10^{-4}) \div (6.25 \times 10^{-3})$ and write the answer in scientific notation.		2

Question 2 (17 Marks)	Commence a NEW page.	Marks
(a) Is $AB \parallel CD$? Give full reasons.		2



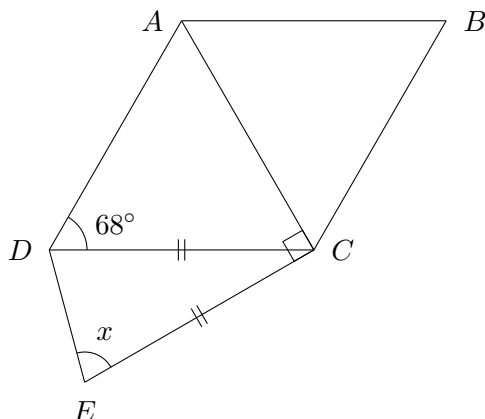
(b) Find the value of x . No reasons required.	2
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- (c) $ABCD$ is a rhombus, where $DC = EC$. $\angle ADC = 68^\circ$ and $\angle ACE = 90^\circ$.

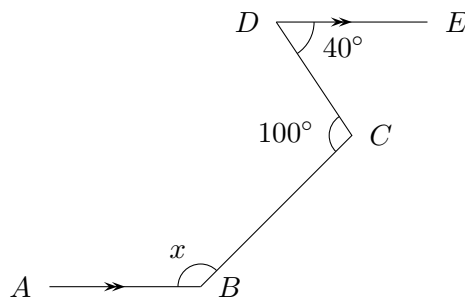
4

Find the value of x , showing all working.



- (d) Find the value of x as shown in the diagram, showing all working.

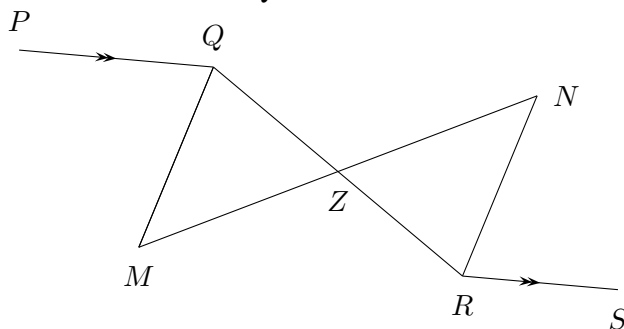
2



- (e) In the given diagram,

- $PQ \parallel RS$
- MQ bisects $\angle PQR$
- NR bisects $\angle QRS$

so that $\angle MQZ = \angle NRZ$ and $MQ = NR$.



- i. Copy the diagram into your writing paper and mark all information on this diagram.
- ii. Prove that $\triangle QMZ \equiv \triangle RNZ$, clearly stating reasons.
- iii. Why does $QZ = ZR$?

2

4

1

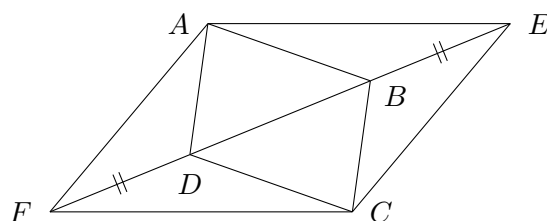
Examination continues on the next page...

- | Question 3 (13 Marks) | Commence a NEW page. | Marks |
|--|------------------------------|-------|
| (a) Fully simplify: | | |
| i. $3\frac{1}{4} : 4\frac{1}{2}$ | | 1 |
| ii. $ab^2 : a^2b$ | | 1 |
| iii. 1 hour : 1 week | | 2 |
| iv. \$5.10 : \$25 | | 2 |
| (b) The ratio of Michael's weight to Linda's is 5 : 4. If Linda weight 58 kg, how much does Michael weigh? | | 2 |
| (c) Find the value of x in simplest form: | $\frac{8}{7} = \frac{21}{x}$ | 2 |
| (d) How much less time does it take Craig, who can type at 60 words per minute, to type a 1 500 word document than Naseem, who types at 45 words per minute? | | 3 |

- | Question 4 (11 Marks) | Commence a NEW page. | Marks |
|--|----------------------|-------|
| (a) Jon receives a 12% pay rise. If his new pay is \$898 per week, calculate his weekly pay before the pay rise. | | 2 |
| (b) If the ratio of cats to guinea pigs is 3 : 5, and the ratio of guinea pigs to dogs is 2 : 3. There are 4 030 animals altogether. | | 3 |

How many dogs are there?

- (c) $ABCD$ is a parallelogram and $BE = DF$.



- | | |
|---|---|
| i. Show that $\angle ABE = \angle FDC$. | 2 |
| ii. Prove $\triangle ABE \equiv \triangle FDC$. | 2 |
| iii. Hence or otherwise, show that $AE = FC$. | 1 |
| iv. Hence or otherwise, show that $AE \parallel FC$. | 1 |

End of paper.

Suggested solutions

Question 1 (12 marks)

- (a) i. (1 mark) – 19
 ii. (1 mark) – 19.45

(b) (2 marks)

- ✓ [1] for correct magnitude.
 ✓ [1] for correct rounding.

$$\sqrt[3]{13^2 - (12.8)^2} = 1.7280245 \dots$$

$$= 1.73 \text{ (2 d.p.)}$$

(c) (2 marks)

- ✓ [1] for correct formula and substitution.
 ✓ [1] for final answer.

$$I = Prn$$

$$= \$850 \times 0.12 \times 5$$

$$= \$510$$

(d) (1 mark)

$$\frac{360^\circ}{8} = 45^\circ$$

(e) (2 marks)

- ✓ [1] for correct substitution.
 ✓ [1] for final answer.

$$\sqrt{\frac{500}{\pi \times 12}} = 3.64$$

(f) (1 mark) - $2.15 \times 10^{-4} = 0.000\ 215$

(g) (2 marks)

- ✓ [1] for correct magnitude of number.
 ✓ [1] for correct magnitude of 10.

$$(4.3 \times 10^{-4}) \div (6.25 \times 10^{-3}) = 6.88 \times 10^{-2}$$

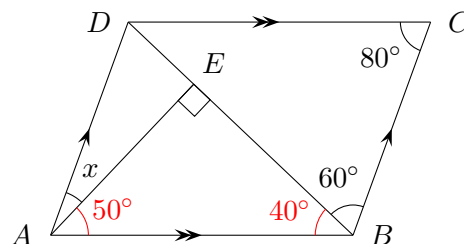
Question 2 (17 marks)

(a) (2 marks)

Yes, $AB \parallel CD$, as the two angles are alternate and equal

(b) (2 marks)

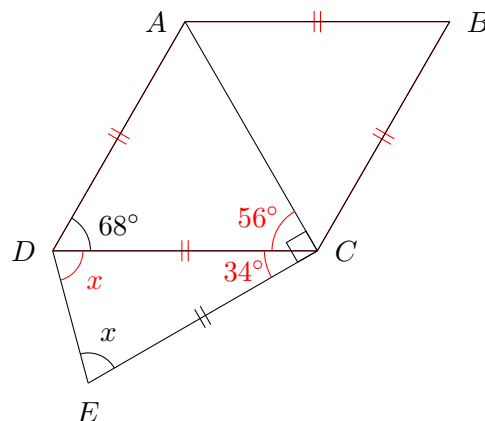
- ✓ [1] for correct working (shown on diagram or otherwise)
 ✓ [1] for final answer.



$$x = 30^\circ$$

(c) (4 marks)

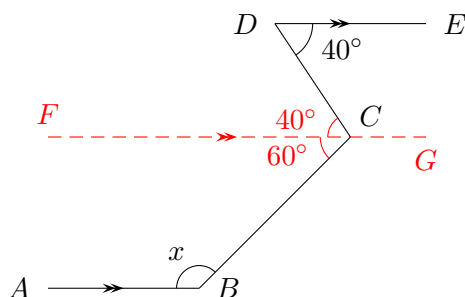
- ✓ [1] for each reason as shown.



- $\triangle ADC$ is isosceles
(as AD and DC are sides of a rhombus)
- $\therefore \angle ACD = \frac{180^\circ - 68^\circ}{2} = 56^\circ$
(base $\angle$ of isosceles $\triangle$)
- $\therefore \angle DCE = 90^\circ - 56^\circ = 34^\circ$
- $\therefore x = \frac{180^\circ - 34^\circ}{2} = 73^\circ$
(base $\angle$ of isosceles $\triangle$)

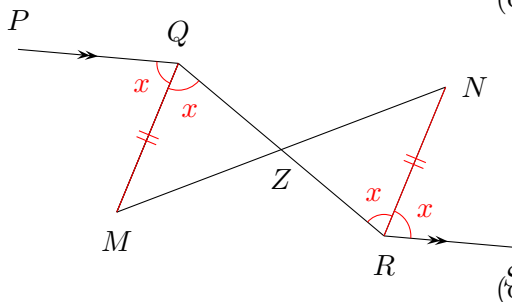
(d) (2 marks)

- ✓ [1] for drawing line FG through C .
- ✓ [1] for final answer, with working.



- $\angle DCF = 40^\circ$
(alternate $\angle$, $DE \parallel FG$)
- $\angle FCB = 60^\circ$
- $\therefore x = 120^\circ$ (cointerior $\angle$ are supplementary, $FG \parallel AB$)

(e) i. (2 marks)



ii. (4 marks)

- ✓ [1] for each correct line shown.

In $\triangle MQZ$ and $\triangle ZRN$:

- $\angle MQR = \angle ZRN$ (given)
- $\angle QZM = \angle NZR$
(vertically opposite)
- $MQ = RN$ (given)

 $\therefore \triangle QMZ \equiv \triangle ZNR$ (AAS)

iii. (1 mark)

$QZ = RZ$ (corresponding sides,
 $\triangle MQZ \equiv \triangle ZRN$)

Question 3 (13 marks)

(a) i. (1 mark)

$$3\frac{1}{4} : 4\frac{1}{2} = \frac{13}{4} : \frac{9}{2} = 13 : 8$$

ii. (1 mark)

$$ab^2 : a^2b = \frac{\cancel{a}b^2}{a^2\cancel{b}} = \frac{b}{a} = b : a$$

iii. (2 marks)

$$\begin{aligned} 1 \text{ hr} : 1 \text{ wk} &= 1 \text{ hr} : 1 \times 7 \times 24 \text{ hr} \\ &= 1 : 168 \end{aligned}$$

iv. (2 marks)

$$\$5.1 : \$25 = \$51 : \$250$$

(b) (2 marks)

$$\begin{aligned} \frac{\text{Michael}}{\text{Linda}} &= \frac{5}{4} = \frac{M}{58} \\ \therefore M &= \frac{5}{4} \times 58 = \frac{290}{4} = 72.5 \text{ kg} \end{aligned}$$

(c) (2 marks)

$$\begin{aligned} \frac{8}{7} &= \frac{21}{x} \\ 8x &= 21 \times 7 = 147 \\ \therefore x &= \frac{147}{8} \end{aligned}$$

(d) (3 marks)

- ✓ [1] each for Craig/Naseem's time to type 1 500 words
- ✓ [1] for final answer.

• Craig:

$$\begin{aligned} \frac{60 \text{ words}}{1 \text{ min}} &= \frac{1 \text{ 500 words}}{x \text{ min}} \\ x &= \frac{1 \text{ 500}}{60} = 25 \text{ min} \end{aligned}$$

• Naseem:

$$\begin{aligned} \frac{45 \text{ words}}{1 \text{ min}} &= \frac{1 \text{ 500 words}}{y \text{ min}} \\ y &= \frac{1 \text{ 500}}{45} = 33.33 \text{ min} \end{aligned}$$

Hence Craig takes 8 min 20 sec less.

Question 4 (11 marks)

(c) i. (2 marks)

(a) (2 marks)

- ✓ [1] for setting up equation/finding 112%
- ✓ [1] for final answer.
- ✓ [-1] for writing $112\% = 898$. $1.12 \neq 898$ and never will!

$$1.12x = 898$$

$$\therefore x = \frac{898}{1.12} = \$801.79$$

(b) (3 marks)

- ✓ [1] for setting up ratio between $C : P : D$.
- ✓ [1] for correct number of animals per part.
- ✓ [1] for final answer.

$$C : P = 3 : 5 = 6 : 10$$

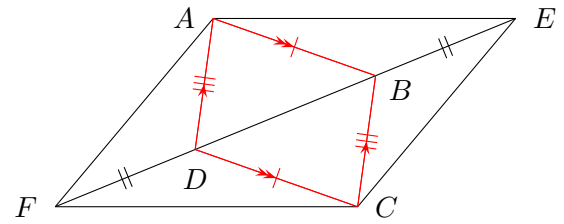
$$P : D = 2 : 3 = 10 : 15$$

$$\therefore C : P : D = 6 : 10 : 15$$

31 parts in total. Find the # of animals per part:

$$\frac{4030}{31} = 130 \text{ animals/part}$$

$$\therefore D = 130 \times 15 = 1\,950$$



- $\angle ABD = \angle BDC$
(alternate $\angle$, $AB \parallel CD$)
- $\therefore \angle ABE = \angle FDC$
($\angle ABE$ and $\angle FDC$ supplementary to $\angle ABD$ and $\angle BDC$ respectively).

ii. (2 marks)

- ✓ [1] for all reasons given to show SAS .
- ✓ [1] for final congruence statement.

In $\triangle ABE$ and $\triangle FDC$,

- $BE = FD$ (given)
 - $\angle ABE = \angle FDC$ (previously proven)
 - $AB = CD$ (opposite sides of parallelogram $ABCD$)
- $$\therefore \triangle ABE \equiv \triangle FDC \text{ (SAS)}$$

iii. (1 mark)

$AE = FC$ (corresponding sides of $\triangle ABE \equiv \triangle FDC$)

iv. (1 mark)

- $\angle DFC$ is alternate to $\angle AEB$
- $\angle DFC = \angle AEB$ (corresponding $\angle$ of $\triangle ABE \equiv \triangle FDC$)
- Hence $AE \parallel FC$ (alternate angles that are equal must stand on parallel lines)